Electrifying School Bus Fleet: A Data-Driven Transition Plan

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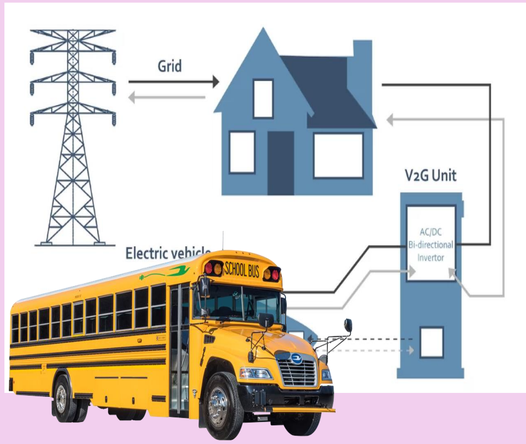
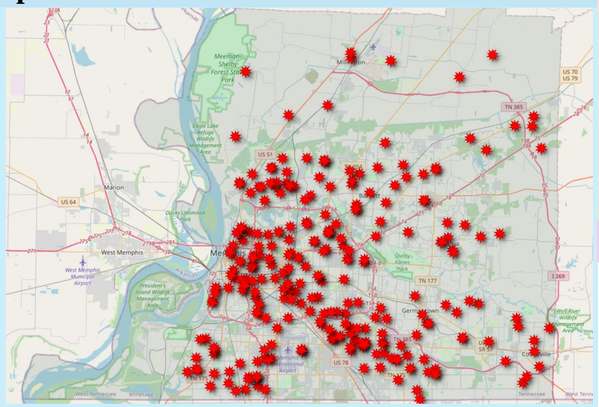
Why Electrify School Buses?

- Large Untapped Energy Resource
  - 480,000 school buses in the U.S. = 60+ GWh of decentralized energy storage
  - High-capacity batteries (100–200 kWh per bus)
- Predictable Use Pattern
  - Fixed schedules allow strategic charging
  - Long idle times enable energy discharge back to the grid (V2G)
- Environmental & Economic Benefits
  - Reduce emissions, fuel dependency, and long-term operating costs
- Grid Integration Potential
  - Smart charging during off-peak hours improves grid stability



Project Scope and Goals

- **Objective:**
  Develop a data-driven V2G electrification strategy for Memphis school buses that:
  - Minimizes costs
  - Optimizes grid integration
  - Supports operational efficiency
- **Scope:**
  - 373 school buses
  - 4 depots: Getwell, E. Brooks Rd., Farmville, Appling City Cove
  - 27,158 daily miles serving 23,500+ students
  - Partnership with First Student, MLGW, & Memphis-Shelby County Schools



Schools in Shelby County

Project Tasks & Timeline

Year 1

Task 1: Charging Strategy Optimization
- Charging Schedule Optimization
- Peak Grid Load Analysis

Year 1

Task 2: Simulation Model Development
- Scenario Testing
- Grid Stability Analysis

Year 2

Task 3: Operational Data Collection
- Charging Pattern Monitoring
- Energy Usage Assessment

Task 4: Prototype Refinement & Optimization
- Charging Schedule Refinement
- V2G Feedback Enhancement

Year 2

Task 5: Full-Fleet Electrification Strategy
- Power Demand Forecasting
- Grid Load Management Solutions

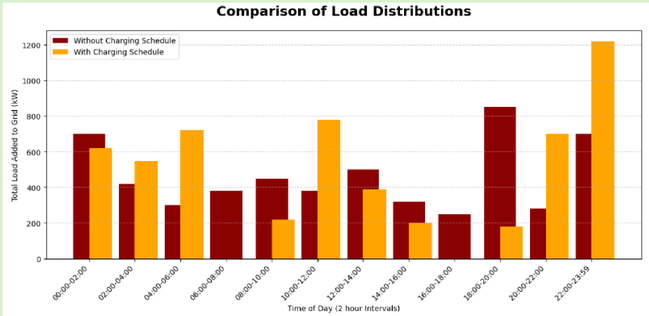
Year 2

Task 6: Detailed Cost Analysis
- Infrastructure Cost Assessment
- Operational Cost & Efficiency Analysis

Year 2

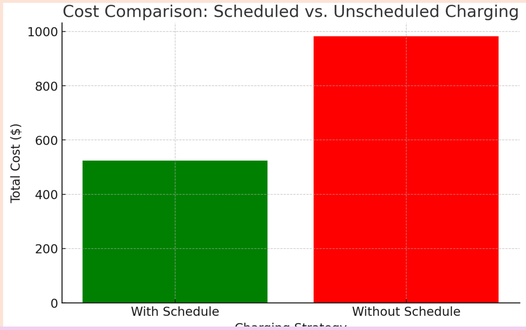
Charging Challenges & Solutions

- **Without Optimization**
- Peak-hour congestion = high costs
- Grid strain from clustered charging
- Underutilized infrastructure
- **With Smart Scheduling**
- 30–40% cost reduction
- Majority charging between 00:00–06:00 & 22:00–00:00
- Avoids peak grid hours (16:00–20:00)
- Balanced load improves grid health



Outcomes & Impact

- **Energy Efficiency:** Smart scheduling saves 30–40% in daily charging costs
- **Grid-Friendly:** Off-peak charging improves stability, reduces infrastructure stress
- **Scalability:** Framework can be expanded to other fleets and cities
- **Sustainability:** Supports cleaner air and lower emissions in school communities

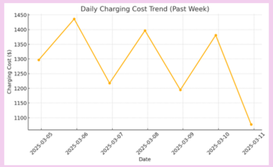


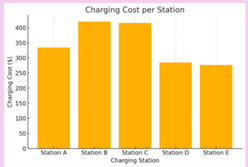
Simulation & Dashboard Development

Overview

Total Buses	50
Total Charging Stations	10
Total Energy Required Today	5,000KWh
Total Cost for Todays Charging	1,200$
Total Charging Slots Available	10

Cost Analysis





Live Charging Schedule

Bus ID	Current Battery %	Charging Station	Scheduled Time	Charging Duration (hrs)	Cost ($)	Status
Bus 101	20%	Station A	10:00 PM	2	$10	✅ Scheduled
Bus 102	15%	Station B	11:30 PM	3	$15	✅ Scheduled
Bus 103	30%	Station C	6:00 AM	1.5	$8	⚠ Pending
Bus 104	5%	❌ No Slot Found	❌ N/A	❌ N/A	❌ N/A	🚨 Warning!

Live Charging Slot Availability

Station Name	Total Slots	Available Slots	Occupied Slots
Station A	1	0	1
Station B	1	0	1
Station C	1	1	0
:: ::			
Station J	1	0	1

Alerts & Warnings

- "2 Buses Could Not Be Scheduled Due to Full Stations!"
- "5 Buses are charging in Peak Hours – Consider Adjusting Schedules!"

User Controls & Scheduling Adjustments

- Recalculate Schedule
- Download Schedule
- Adjust Constraints

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